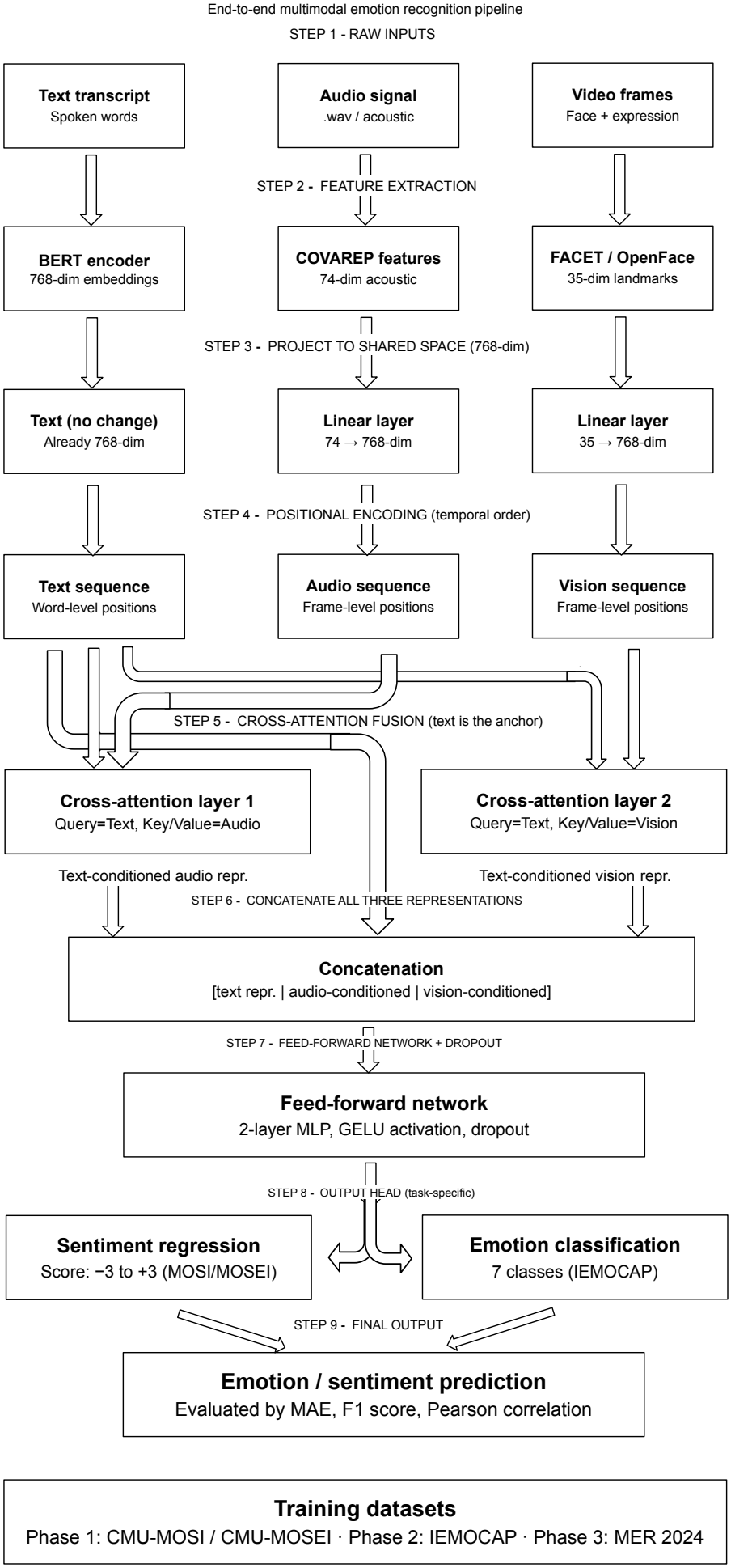

Text-Anchored Cross-Attention Transformer
End-to-end multimodal emotion recognition pipeline
STEP 1 - RAW INPUTS
Text transcript
Spoken words
Audio signal
.wav / acoustic
Video frames
Face + expression
STEP 2 - FEATURE EXTRACTION
BERT encoder
768-dim embeddings
COVAREP features
74-dim acoustic
FACET / OpenFace
35-dim landmarks
STEP 3 - PROJECT TO SHARED SPACE (768-dim)
Text (no change)
Already 768-dim
Linear layer
74 → 768-dim
Linear layer
35 → 768-dim
STEP 4 - POSITIONAL ENCODING (temporal order)
Text sequence
Word-level positions
Audio sequence
Frame-level positions
Vision sequence
Frame-level positions
STEP 5 - CROSS-ATTENTION FUSION (text is the anchor)
Cross-attention layer 1
Query=Text, Key/Value=Audio
Cross-attention layer 2
Query=Text, Key/Value=Vision
Text-conditioned audio repr.
Text-conditioned vision repr.
STEP 6 - CONCATENATE ALL THREE REPRESENTATIONS
Concatenation
[text repr. | audio-conditioned | vision-conditioned]
STEP 7 - FEED-FORWARD NETWORK + DROPOUT
Feed-forward network
2-layer MLP, GELU activation, dropout
STEP 8 - OUTPUT HEAD (task-specific)
Sentiment regression
Score: −3 to +3 (MOSI/MOSEI)
Emotion classification
7 classes (IEMOCAP)
STEP 9 - FINAL OUTPUT
Emotion / sentiment prediction
Evaluated by MAE, F1 score, Pearson correlation
Training datasets
Phase 1: CMU-MOSI / CMU-MOSEI · Phase 2: IEMOCAP · Phase 3: MER 2024

Experimentation Plan

To test if the Text-Anchored Cross-Attention model actually works better, I plan to run these four phases of experiments:

1. Baseline Comparison (Which fusion method is best?) I will build all three models and compare their accuracy (MAE and F1 score) on the CMU-MOSI dataset:

- **Early Fusion:** Smashing all features together at the start.
- **Late Fusion:** Training 3 separate models and averaging the score.
- **My Proposed Model:** Using Cross-Attention.

2. Ablation Study (Does adding audio/video actually help?) I will strip the model down and test it in pieces to prove that we need all three modalities:

- Test 1: Text only (Just BERT).
- Test 2: Text + Audio.
- Test 3: Text + Vision.
- Test 4: The full model (Text + Audio + Vision).

3. Tuning the Model I will experiment with different settings to get the best accuracy:

- Testing different numbers of attention heads (4 vs 8 vs 16).
- Trying different dropout rates (0.1 to 0.5) to stop it from overfitting.

4. Testing on Other Datasets Once it works on CMU-MOSI, I will test if the model can handle other types of data by fine-tuning it on:

- IEMOCAP (for 7-class emotion classification).
- MER 2024 (to see if it works on "in-the-wild" video clips).